Physical Security

Definition of Terms

Security

- Is a predictable state or condition
- Freedom from fear or danger
- Where one can pursue his objectives

Physical Security

- A system of barriers placed between the adversary and the subject of protection

The 3 categories of Security

- Physical
- Information
- Personnel

Foundation of Physical Security

1. System of ‘Defense in Depth’, called ‘compartmentalization’
2. Accumulated Delay Time

Defense In-depth

It is generally accepted that there are 3 layers of physical security:

1. The outer perimeter barriers located at the outer edge of the property referred to as the first line of defense.

2. The exterior walls, ceilings, roof, and floors of the building the second line.

3. Temporary, or partitioned areas within the building known as the third line of defense.

Defense in Depth

Perimeter barrier

(1st Line of Defense)

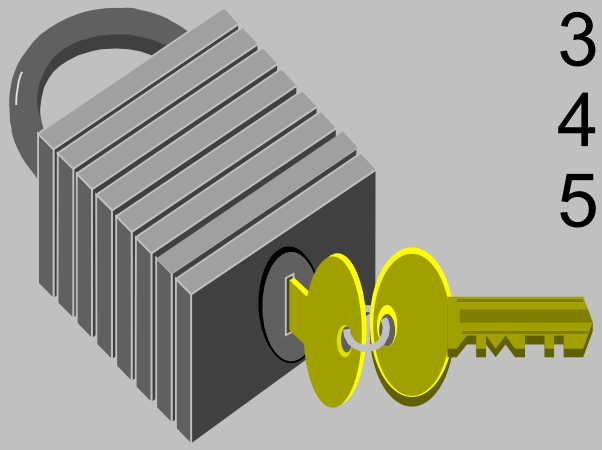
Main purpose

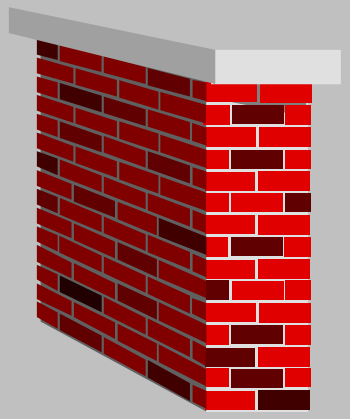
To deny or impede access or exit of unauthorized person/s.

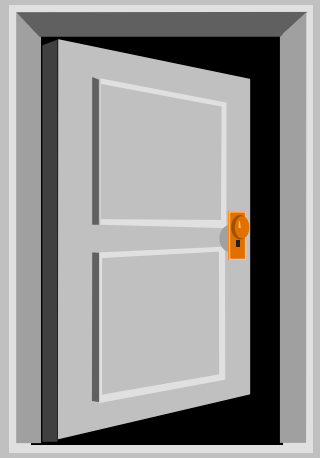
Building exteriors

(2nd Line of Defense)

1. Wall
2. Doors
3. Windows
4. Roof Openings
5. Fire Escapes







Interior controls

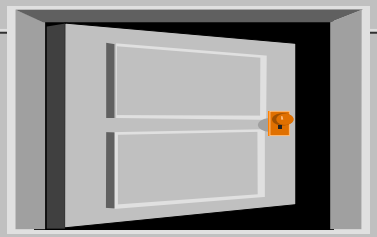
(3rd Line of Defense)

1. ID System
2. Entry
3. Exit
4. Restricted Area
5. Alarm Sensors
6. Communication Systems

   a. Local Tel Exchange
   b. Two-way Radio
   c. Commercial Tel Svc
   d. Call Boxes
   e. Intercommunications
   f. Paging & Recall Systems

7. Guards

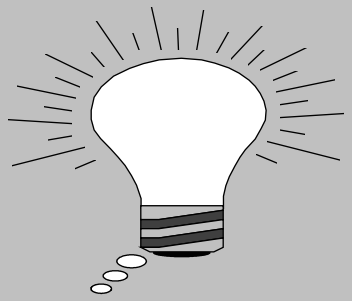
Physical Barriers

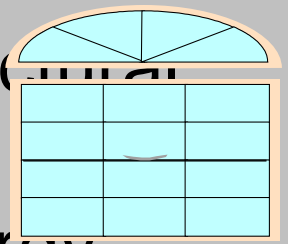


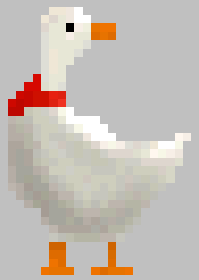
5 TYPES:

* Natural

* Animal
* Human
* Structural







* Energy

Barriers serve three basic functions:

- Delay
- Psychological deterrent
- Supplement and/or compliment the on-site security personnel needs

Positive barriers should be set up to:

- Control vehicle and pedestrian traffic
- Enable identification of people arriving and departing
- Provide a buffer zone for more sensitive areas of the company

Purpose of perimeter barriers

- Defines the boundary of the property to be secured
- Creates a physical and psychological deterrent to unauthorized entry
- Delays intrusion, thus facilitating apprehension of intruders
- Assists in a more efficient and economical employment of guards
- Facilitates and improves the control of pedestrian and vehicular traffic

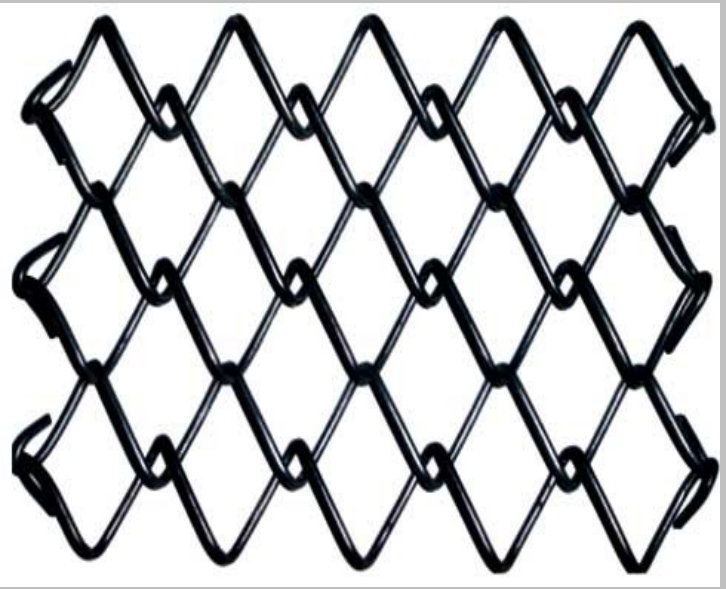
Types of fencing

1. Solid
2. Full-view

Types of fencing materials authorized for the protection of ‘restricted’ areas.

- Chain link
- Barbed Wire
- Concertina
- Barbed Tape

Chain Link



Chain Link Fencing

A major advantage of Chain Link fencing is that it provides good visibility from both sides, for both guards and police patrols.

The disadvantages are that it creates an institution-like appearance to the organization.

Specifications for Chain Link

1. Must be a minimum of 7’ in height
2. 9 gauge or heavier
3. 2” mesh openings
4. Have a twisted and barbed salvage at top and bottom
5. Securely fastened to metal or concrete posts set in concrete
6. The fencing material must reach within 2” of hard ground or paving
7. On soft ground be buried to compensate for shifting soil and sand

8. All bolts and nuts holding hardware attachment on the fence should be welded

9. Gates in the fence-line should be as high as the fence

10. ‘Top guard’ for the gate should be vertical

Barbed-Wire Fencing

Barbed-Wire Fencing

1. Twisted, double-strand
2. 12 gauge
3. With 4 point barbs spaced equally
4. Not less than 7’ (excluding topping)
5. Fixed firmly to post not more than 6’ apart
6. Distance between strands not to exceed 6” with a vertically interlaced wire at the mid-way between posts.

Concertina Wire Fencing

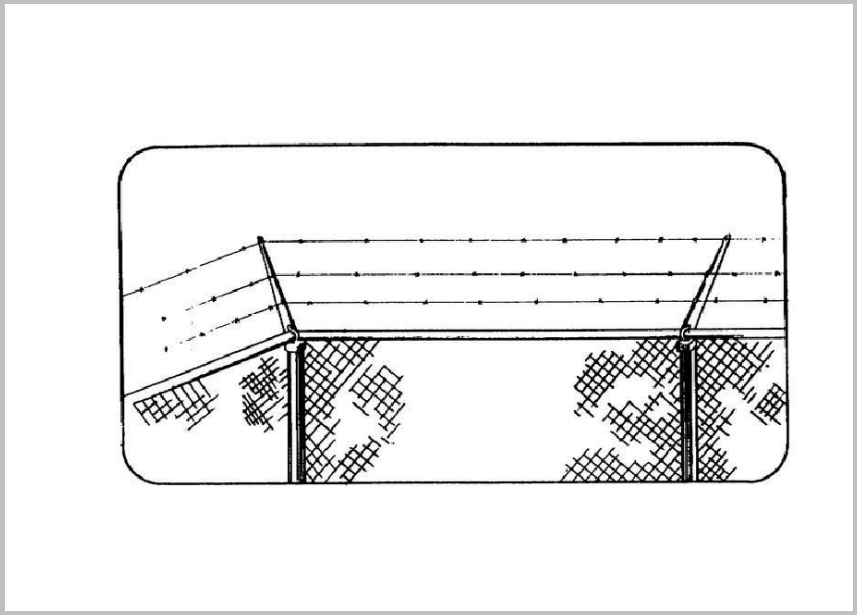
Concertina Wire Fencing

1. Concertina barbed wire or tape to be made from high-strength steel clipped together to form a cylinder.

2. Opening to 50’ and 3’ in diameter.

3. Barbed tape should have a breaking strength of 500lbs, with an overall width of ¾”, with 7/16” barbs spaced at ½” intervals along each side.

Fencing Top-Guard



Fencing Top-Guard

1. Top guard is the overhang of barbed wire or tape along the top of the fence, facing outward and upward at approx a 45 degree angle.

2. The supporting arms must be permanently fastened to the tops of the posts to increase the overall vertical height by 1’.

3. Three strands of barbed wire, spaced 6” apart, must be installed on the supporting arms.

Utility Openings

1. Utility openings of 10” or more in diameter must be secured to prevent unauthorized opening.

2. Drainage ditches, culverts ducts, and other opening of more than 96 sq” cross sectional area should be secured by welded bar grills.

Clear Zones

- Clear zones of 20 feet or more should exist between the perimeter barrier and exterior structures, parking areas and natural or man-made features.

- A clear zone of 50’ or more should exist between the perimeter barrier and the structures within the protected area except where a building wall forms part of the perimeter barrier.

Signage

- Signs are advisable spaced every 100’ either on the fence or in front, stating that the property is not open to the public and intruders will be prosecuted.

- This act as a deterrent.

Doors

1. Fire safety doors required by safety regulations are generally undesirable to security.

2. The usual method of opening fire doors is by pushing against a panic bar installed at the mid-rail position on the secure side of the door.

3. Security should be enhanced at fire doors by the use of audible alarms.

4. Unusually long warehouse doors should be padlocked inside at both ends.

5. Hinge pins exposed to the outside must be protected by either spot welding, or hinge bolts set into the door frame.

Windows

Types of protection available include:

- ❑ burglar resistant glass
- ❑ protective iron or steel bars
- ❑ good grade heavy steel mesh
- ❑ chain-link fencing material

- Experts advise that exterior windows with a ledge height of more than 18’ are seldom used by intruders.

- Windows less than 18’ above ground or less than 14’ from the trees, poles, or adjoining buildings and larger than 96sq” in area, should be protected.

4. Over 50% of all break-ins are through windows.

5. Glass

a) Types:

i) Standard Plate Float

ii) Tempered (toughened)

iii) Wired Glass (External/internal use as required by fire/safety codes. Shatter and fragmentation resistant.)

iv) Laminated Safety Glass

6. Plastics

a) Types:

I) Standard Acrylic, 17 x the impact resistance over glass of the same thickness, and 50% lighter.

II) Polycarbonate, 30 x stronger than acrylic of the same size.

b) Spalling

The property built into safety glass which allows it to chip or fracture into harmless particles rather than in large pieces which are dangerous.

7. Types of Burglar-Resistant glass:

   a) UL ‘Safety Glass’

   b) Plastic glazing sold under the trade names of ‘Tuffak’ Rohm and Has

8. Underwriter laboratories mark products with a ‘UL’ symbol indicating they met certain recognized standards following laboratory tests.

9. Miscellaneous

Some special security problems are generated by the following features that breach the perimeter barrier:

- Side walk elevators
- Utility tunnels
- Operational tunnels
- Storm sewers
- Storm doors
- Piers, docks and wharves.

Security Alarms

The purpose of Intrusion Detection Systems (IDS) is:

To economize

To substitute in place of other security measures

To supplement by providing additional controls

The basic principles of operation are:

- Breaking an electrical circuit
- Interrupting a light or infra-red beam
- Detecting sound
- Detecting vibration
- Detecting sources of heat
- Detecting a change in capacitance due to the penetration of an electrostatic or electromagnetic field.

Security alarms

Security alarms

Alarm systems fall into two main categories.

a) Electro-Mechanical

Metal foil
Magnetic switches
Wire service detection
Pressure mats
Ribbons or wafers

b) Volumetric

Capacitance

Vibration

Microwave

Ultrasonic

Passive infra-red

Photo-electric

5. Proximity and capacitive sensors use essentially same principle.

6. Dual-tech sensors are combination or either IR and M-wave or IR and U-sonic.

7. Checklist for Indoor Sensors:

   - Use in stable environment
   - Microwave warning
   - PIR/Ultrasonic cautious
   - PIR/View

8. Outdoor Intruder Protection

Fence or Wall Mounted

a. Vibration detectors
b. Inertia detectors
c. Micro phonic
d. Capacitance(E-field)
e. Fibre Optic

Buried Sensors

a. Leaky Cable
b. Balanced pressure system
c. Capacitance

d. Acoustic

e. Fiber Optics

Free Standing Sensors

a. Infra-red beams

b. Microwave (beams or single)

c. Electric fencing

d. CCTV Video Motion Detection

e. Photo-electric systems

Line Supervision for Security Alarms

Line Supervision is very important, as it is the weakest link in IDS system. Line needs tamper detection. Methods of supervision are:

a. dedicated lines

b. proprietary lines

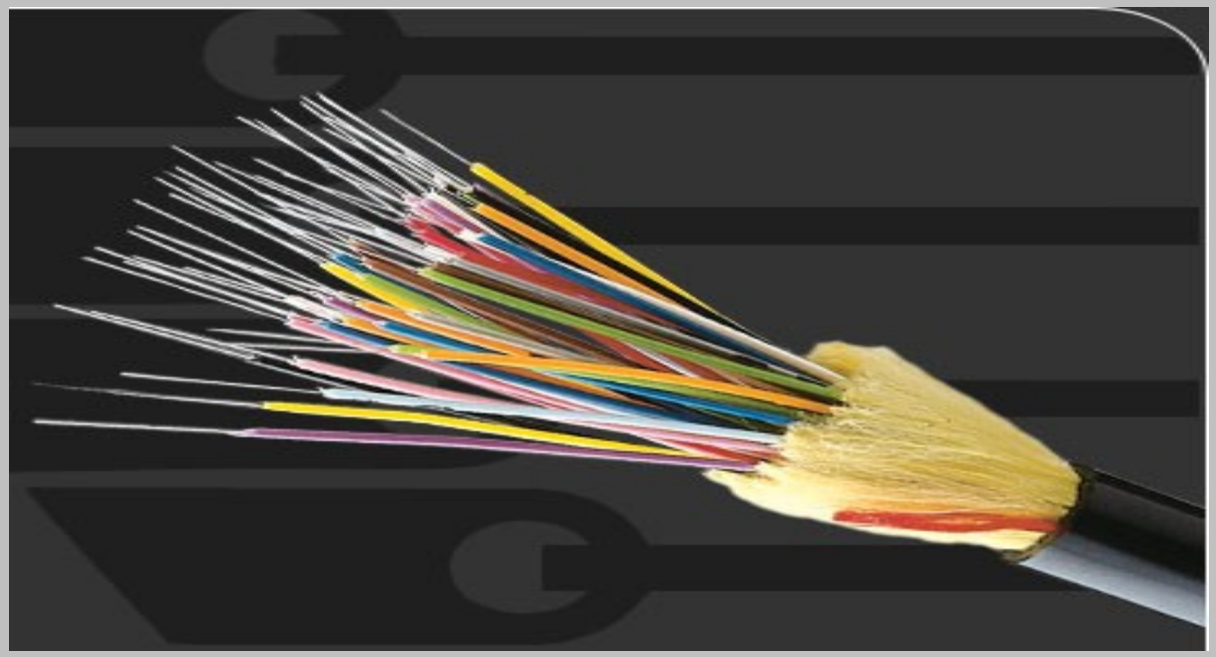
Line should always be buried when it goes outside of building, and inside should always be inside conduit. The line circuits should be able to be checked automatically by the system.

Security Communications

Wire transmission of alarm and audio signals.

- Alarm Signals
- unshielded pair of conductors, cannot transmit audio over these.
- Audio Transmissions Shielded twisted pair for audio.
- Video Signal coaxial or fiber optics. ISDN video over phone lines to include color.

Fiber Optics Versatile to cover all above requirements. Note power cannot be transmitted over fiber optic carrier.



Fibre Optic Technology

Fiber walls made of glass and organic polymers, almost impossible to access information, is increasingly used to link computers, CCTV, and command modules in high risk area.

Advantages are:

1. Low level of false alarms
2. Compact
3. Immune to electrical or electro-magnetic disturbances
4. Immune to wind born litter, animals, falling leaves, snow or weather changes.
5. Will not over heat, short circuit or spark.

Methods of Wire Transmission

Loop System

Number of devices installed at intervals on a pair of twisted wires looped throughout an area to a central control point.

Point-to-Point System

Each sensor is connected directly to a control center by a pair of wires.

Multiplexing

Several simultaneous messages on the same circuit. Defined as time sharing of dedicated voice grade channels.

Types of alarm detection are:

Local Alarm Systems- Protective circuits activate a visual or audible signal in the immediate vicinity of the protected object or area.

Auxiliary System- This system is one in which the installation-owned system is a direct extension of the police and/or fire alarm systems(regarded as the least effective system).

c. **Central Station System**- Transmit alarms to a central station outside the installation from which appropriate action is taken such as notifying the police or fire department.

d. **Proprietary System**- Similar to a central station system except it is owned by and located on the installation, and response to alarms is by the on-site security or fire fighting staff.

There are three basic elements to an IDS -

a. Sensor or Trigger (Detector)

b. Circuit which transmits the change of condition

c. The signaling device, called the Enunciator

The functions of alarm systems are to:

a. detect fire or intrusion

b. emergency notification

c. monitoring equipment or facility conditions.

Between 95-98% of all alarm activations are said to be false. The most common causes of false alarms are:

a. user negligence

b. poor installation or servicing

c. faulty equipment

Glossary of terms for IDS systems

1. **Actuator**- The button, a magnetic switch, or thermostat that will cause the system to alarm.

2. **Annunciato**r- This is the monitor, using visual or audible signaling that indicates the condition of the associated circuits.

3. **Capacitance**- The property of two, or more objects, which enables them to store electrical energy in an electrostatic field between them.

4. **Capacitance Proximity Sensor**- Records the change in capacitance or electrostatic field to detect penetration through windows, ventilators and other openings and can be used to detect attempted penetration into safes or storage cabinets.

5. **Conductor**- A material which transmits electric current, e.g. wire.

6. **Control Unit**- The terminal box for all sensors.

7. **Fail-safe** - A term applied to systems which will indicate by signaling or otherwise when they are incapacitated through component failure, or low performance.

8. **Duress Sensor** - Used to call assistance, consists of hand or foot operated switch.

9. **False Alarm** - An activation of sensors by other than an intruder. (In the US they say by causes unknown).

10. **Intrusion Detection System (IDS) -** The combination of components including sensors, control units, transmission lines and monitor units integrated to operate in a specific manner.

11. **Intrusion Detection Sensor**- Devices that initiate alarm signals by sensing the stimulus, change or condition for which it was designed.

12. **Local Audible Alarm** – An electronic bell or sounder for outdoor or indoor use in the vicinity of the protected area.

13. **Microwave Sensor**- A radio frequency (RF) transceiver having a frequency range in GHZ which detects motion.

14. **Monitor**- A device which sensors and reports the condition of a system.

15. **Motion Sensor**- Detects movement inside the area to be protected.

16. **Passive Ultra-Sonic**- Detects the sounds of forced entry throughout walls, ceilings and doors.

17. **Penetration Sensor**- Detects the entry through walls, windows, doors or any other opening into the protected area.

18. **Photoelectric System**- Consists of two separate units, a transmitter and a receiver. An interruption of the light beam between the two causes an alarm.

19. **Ultrasonic**- The frequency range of sound inaudible to humans. In IDS System, the range is usually between 21.5 KHZ and 26 KHZ.

20. **Ultrasonic Motion Sensor**- Detects by frequency shift the motion of an intruder inside the protected area.

21. **Vibration Detector**- Detects forced entry through metal barriers placed over windows and ventilators or attempts to drill, saw, or cut through walls, ceilings, floors or doors.

Access Control Systems (ACS)

Access card or badge have several types:

a. Magnetic card

b. Optical

c. Hollerith

d. Weigand

e. Proximity

f. Capacitance

g. Smart card

Badge Reader

Interprets card, identifies card number, compares to database, checks authority/access rights and allows or denies holder access.

Electrified Locks

Part of an access control system and has three purposes:

- To unlock door on authority from reader
- To lock/unlock at remote locations, usually back up by CCTV
- Auto lock/unlock on time basis.

Locks provide flexibility with fire alarm system and controls in emergency.

There are three types of electrified locks:

- Electric Strike- most common
- Electromagnetic- power magnet
- Electric bolt- positive action

“Fail Safe” and “Fail Soft” “Fail Secure”:

- **‘Fail Safe’** - On power failure, door unlock for egress
- **‘Fail Soft’** - No recording of entry/exit
- **‘Fail Secure’-** On power failure, door remains locked

ACS Control Units

Remote Control units, that translates communications and perform tasks between card reader, electric door locks, and the central control unit.

Central control units comprise of:

1. Enrollment console
2. Central processor
3. Printer
4. Multiplexer

Biometrics

They all use a **Personal Identity Verifier**(PIV), this is the unique pattern or other feature of a person to be verified.

Types include:

- Hand geometry
- Iris
- Retina
- Signature
- Voice
- Vein Pattern
- Fingerprint

Advantages and disadvantages of Biometric Systems are that some are intrusive, some easy to forge, and some unreliable, however developments continue to improve on their accuracy and reduced error rates.

Biometrics

Other advantages are that they are measure a unique trait that does not change. There is nothing to lose or forget, and can integrate with other industrial standard readers.

Error rates are classified into two types:

Type 1- erroneous rejection of authorized users.

Type 2- false acceptance of unauthorized users.

General Access Control Note

a. Visiting lorry drivers must have a separate waiting rooms.

b. Security classifications should never appear on ID badges

c. Weakest point in a personnel identification system is the failure of enforcement by security guards.

Protective Lighting

Protective lighting provides a means of continuing, during the hours of darkness, a degree of protection maintained during daylight hours.

The importance is three fold:

a. It serves as a deterrent to intruders

b. It assists the security force

c. It serves as an essential element of an integrated physical security programmed.

Protective Lighting

The general characteristics of protective lighting are:

a. It is relatively inexpensive to maintain

b. It will probably reduce the number of security forces personnel,

c. It provides protection for them by reducing the element of surprise by the intruder.

d. It requires less intensity than working light.

Planning considerations for installation & maintenance of protective lighting

1. Consider the effects of the local weather conditions.

2. Fluctuation of voltage in the power source.

3. Standby power source capable of maintaining the desired level of lighting, particularly in all vital areas.

4. Establish a ledger to maintain a burning time record based on the life expectancy of the lamp.

Limited and Exclusion Area Needs:

1. These areas must have protective lighting on a permanent basis during the hours of darkness, at both the perimeter and access control points.

2. The wiring circuit should be such that the failure of one or more lights should not affect the operation of the remaining lights.

3. The control for such lighting should be by security.

4. The lighting must be positioned so as to prevent glare or silhouetting the security staff.

5. The cone of light should be directed downward and away from structures or the areas protected, and away from security personnel assigned to such protection.

6. At perimeter barriers the lighting should be located sufficiently within the protected area and above the fence so that the light pattern will cover a ground area.

7. Light levels recommended for these areas are:

a. 0.2 foot-candles, measured horizontally at 6” above ground and outward from the area perimeter barrier to at least 30’

b. The lighting level inside the exclusion area, and on structures containing nuclear weapons, must be sufficient to detect persons in the area or at entrances to any structures.

c. Levels at access control points should be sufficient to enable comparison of ID cards with the holder.

d. Protective lighting should never be relied on alone but should be used with other security measures.

General Principles of Protective Lighting

Protective lighting should enable security staff to see without being seen, with adequate and even levels of light over bordering areas, where possible directed to glare in the eyes of the intruder, and with little light on actual patrol routes.

To be effective, protective lighting should act as deterrent making detection likely.

Types of Lighting

There are four main categories of lighting:

Continuous - is the most common protective lighting, consisting of stationary luminaries, and is generally employed for glare projection and control lighting.

Standby - Luminaries are stationary, but are not continuously lit.

Moveable - Consists of manually operated movable search light which may be lit during darkness or as necessary, and they are usually used to supplement ’Standby’ or ‘Continuous’ lighting

Emergency - The system may duplicate any of the other three (3) systems in whole or part, and use generally limited to times of power failure or other emergencies, as they rely on alternative power sources such as portable generator s or batteries.

Piers and Docks

- They should be protected by lighting both on water approaches and at the pier area.

- Decks on open piers should be lit to at least 1.0 foot-candles. Water approaches, extending to a distance of 100’ from the pier should be lit to at least 0.5 foot-candles.

- The US coastguards should be consulted for approval of proposed protective lighting adjacent to navigable waters.

3. **Wiring Systems**

- Circuits should be arranged so that the failure of any one lamp will not leave a critical or vulnerable position in darkness.
- Feeder lines should be located underground (or sufficiently inside the perimeter in case of overhead wiring) to minimize sabotage or vandalism from outside the perimeter.
- Periodic maintenance inspections should be made of all electrical circuits to replace or repair worn parts.

4. Department of Army Lighting Spec Table (FM 19-30, Mar 79')

Location	Foot-candles
Perimeter of outer area	0.15
Perimeter of restricted area	0.40
Vehicular entrances	1.00
Pedestrian Entrances	2.00
Sensitive inner area	0.15
Sensitive inner structure	1.00
Entrances	0.10
Open Yards inc Car parks	0.20
Decks on open piers	1.00

** Foot-candles on horizontal plane, ground level*

Definitions

a. Candle Power - One candle power is the amount of light emitted by one standard candle.

b. Lumen - One lumen is the amount of light required to light an area of one square foot to one candle power.

c. Foot Candle - One foot-candle equals one lumen of light per square foot of space. The intensity of illumination is measured in foot-candles.

d. 1 lux = 10ft. Candles

Types of Lighting Used for Protective Purposes

Incandescent- A common light bulb, provides instant light on power up, and can be manufactured such that light can be reflected or diffused.

Gaseous Discharge Lamps - There are two main types.

- Mercury Vapour (soft blue-white light)
- Sodium Vapour (gives soft yellow light)

The main disadvantages of gaseous discharge lights are that they take between 2-5 minutes to light (warm-up) and even longer when already hot, i.e. after a power failure- this is called the 'restrike time'.

The advantages are that they are much more efficient than incandescent lamps, widely used where fog is a problem and frequently found on a bridges and highways.

Quartz lamps - They emit a very bright light almost instantaneously when powered up.

They are excellent for use along perimeters and in critical areas, and are frequently used at very high wattage (500-1000 watt), therefore more costly to run and they are more costly to replace.

The individual life expectancy is also much lower than the other two types, although the initial purchase and installation cost is the cheaper of the types covered here.

Types of Lighting Equipment

- **Floodlights** - Directional, with some diffusion to light specific areas.

- **Searchlights** - Lights that project light in a concentrated beam, and can be used as initial response to perimeter area intrusion incidents.

- **Fresnels** - Wide beam units used to extend the illumination in long horizontal strips. They project a narrow beam approximately 180 degrees horizontal and 15 to 30 degrees vertical. On perimeter little vertical light is lost.
- **Street Lights** – Used in parking areas and top-up applications and produced diffused light.

Safes are designated either fire resistive or burglary resistive.

There is no Federal or state law or code mandating the rating of safes, and two organizations currently rate safes in the U.S; the Safe Manufacturers National Association (SMNA), and the Underwriter's Laboratories(UL).

SAFES

- Safes are designated either fire resistive or burglary resistive.

Fire Resistant Safes

Fire resistant safes offer very little protection against the safe-cracker, and once exposed to a fire the original degree of protection offered is lost, and it must be replaced.

Underwriter's Laboratory (UL) Tests

a. Fire exposure
b. Explosion protection
c. Impact, fall of 30'

Safe is heated for 4 hours to 2000F, then left for 4 hours to cool off, then heated up again.

Explosion impact test: Impact test drop of 30'. Return to furnace after cooling.

The following tables, list the categories and ratings:

Fire Resistant Containers UL Record Safe Classifications

Classification	Temperature	Time	Impact
350-4	1700F	4hrs.	Yes
350-2	1850F	2hrs.	Yes
350-1	1900F	1hr.	Yes
350-1*	2000F	1hr	Yes
350-1	2150F	1hr	No

UL Computer Media Storage Classification

150-4	2000F	4hrs.	Yes
150-3	1900F	3hrs	Yes
150-2	1850F	2hrs	Yes
150-1	1700F	1hr	Yes

UL Insulated Vault Door Classification

350-6	2150F	6hrs	No
350-4	2000F	4hrs	No
350-2	1850F	2hrs	No
350-1	1700F	1hr	No

*Insulated Record Container

4. Burglary and Robbery Resistive

Burglary and Robbery resistive mercantile safes are classified by SMNA specs, by UL ratings and by design features of doors, wall and lock. A classification table in this regard is set out below:

UL Money Safe Classifications

Class	Description		Construction
TL-15	Tool Resistant	Weight	At least 750lbs., or anchored. At least 1”thick steel, or equal
		Attack	Door and front face must resist with common hand and electric tools for 15mins.

TL-30	Tool Resistant	Weight	At least 750lbs., or anchored. At least 1”thick steel, or equal.
		Attack	Door and front face must resist attack with common hand and electric tools plus abrasive cutting wheels and power saws for 30 minutes
TRTL 30 listed	Tool and Touch Resistant	Weight	At least 750lbs.
		Attack	Door and front face must resist, attack with tools above, and oxy fuel gas cutting or welding torches for 30 minutes.

UL Money Safe Classifications (Continued)

Class	Description		Construction
TRT-30x6	Tool and Torch resistant	Weight Attack	At least 750 lbs. Doors and entire safe body must resist attack with tools and torches listed above, plus electric impact hammers for 30mins.

TXTL-60	Torch, tool and explosive resistant	Weight Attack	At least 1000 lbs. Door and entire safe body must resist, attack with tools and torches listed above, plus 8oz. of nitroglycerine or its equal for 60mins.

As of 31st January 1980, UL stopped issuing TRTL-30 labels, replacing it with the TRTL-30X6 label which requires equal protection on all six sides of the safe. Some manufacturers, however, continue to produce safe meetings the TRTL-30 standard in order to supply lower priced containers, which provide moderate protection against tool and torch attack.

The design features of burglary resistant safes generally require the door to be made of steel and at least 1.5” thick.

Safe Cracking Methods

a. Drilling or Punching- Accomplished by knocking off combination dial and drilling hole to expose the locking device.

b. Burning- Using high temperature oxyacetylene torches or “burning bars” to cut an opening in the wall or door of the safe.

c. Peeling- Attacking the seams of the metal plates with pry bars and other tools to peel back layers of metal and thus exposing either locking mechanism or interior.

d. **Ripping**- Similar to peeling except ripping can be accomplished against solid, metal walled container with a thin wall.

e. **X-ray**- Used to reveal the position of the combination and mechanism needed to open the safe.

f. **Explosives**- Nitroglycerin and plastic explosives are still used but not as much as in the past.

g. **Power Tools**- E.g. Rotary devices, hydraulic tools and power drills.

h. **Manipulation**- Very few safe-crackers have the skill to use this technique.

Vaults

•Defined as enlarged safes, usually made of high quality reinforced concrete, except the door.

•The five sides should be twice as thick as the door but never less than 12”

•The vault door is usually made of high grade steel and normally 6” thick.

• Vaults are normally installed at or below ground level because of their heavy weight.

• It is also usual to have a clear sterile walkway around the five above base sides when installed below a buildings foundations.

• Vaults should be fitted with two types of alarm detection devices, capacitance and vibration.

- Ratings for vaults are established by the Insurance Services Office (ISO).
- Fire resistive enclosures. **“No More Than**” (NMT) 12 feet high or in excess of 5000 cu ft.

There are 3 Class Types:

a. Two Hour
b. Four Hour
c. Six Hour

Classification:

a. Burglary Resistive
b. Fire Resistive

Ground Support vs. Structure Support

a. Ground Supported- Reinforced concrete base, (8” for 4 hour, and 10” for 6 hour), independent building with no fire resistance.

b. Structure Supported- Permitted to use existing floor if auto sprinkler system in use.

Protection for Vault Surfaces

a. Vault Doors, Steel, drill and touch resistant, dial combination and time lock.

b. General Service Administration (GSA) Approved Vaults

(The lower the number the greater the security).

Class	Features
6	Surreptitious 30 min Forced attack 0 min
5	Surreptitious 30 min Forced attack 10 min Lock manipulation 20 hrs. Radiology attack 20 hrs.

LOCK AND KEY SYSTEMS

1. The lock is the most accepted and widely used security device.

2. Irrespective of their design and/or cost they should only be regarded as delaying devices.

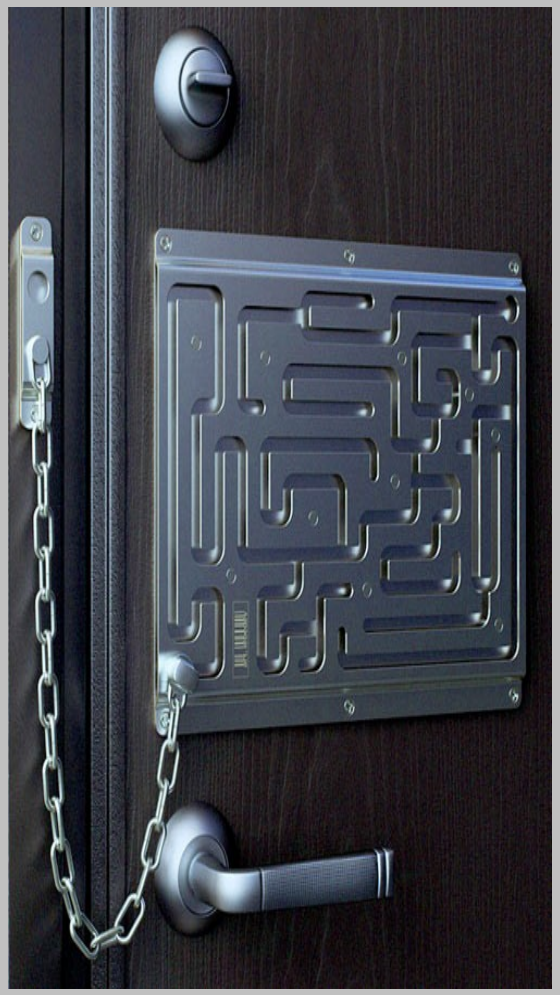
3. Types of Locks by Installation

   a. Mortise lock (or Lockset), includes latch and deadbolt.

   b. Bored or Cylinder lock

   c. Unit lock

   d. Rim lock



Types of Locks by Operation are:

Key Locks - May be picked, possibility of loss or compromise of key.

Warded Locks - offer no security, worst selection for security

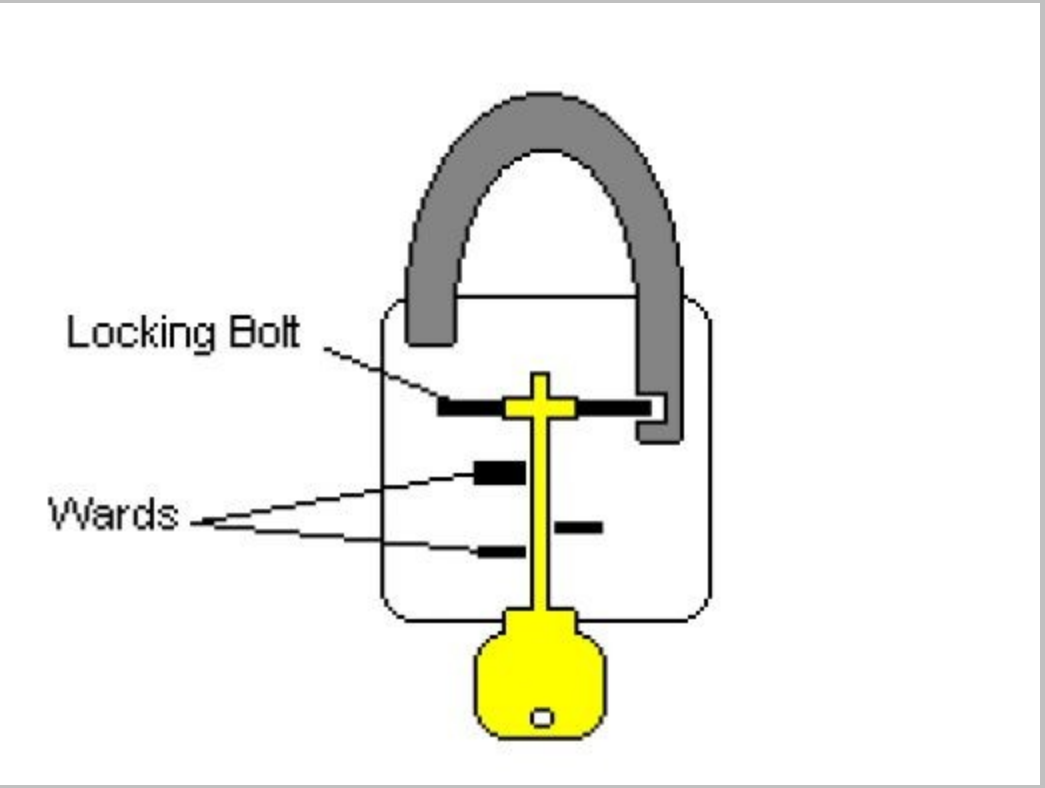
Disc Tumbler Locks - Sometimes called wafer locks, more secure than warded locks, used on cars, desks and cabinets.

Pin Tumbler Locks - Used extensively in commercial, military, and residential security, more secure than previous two.

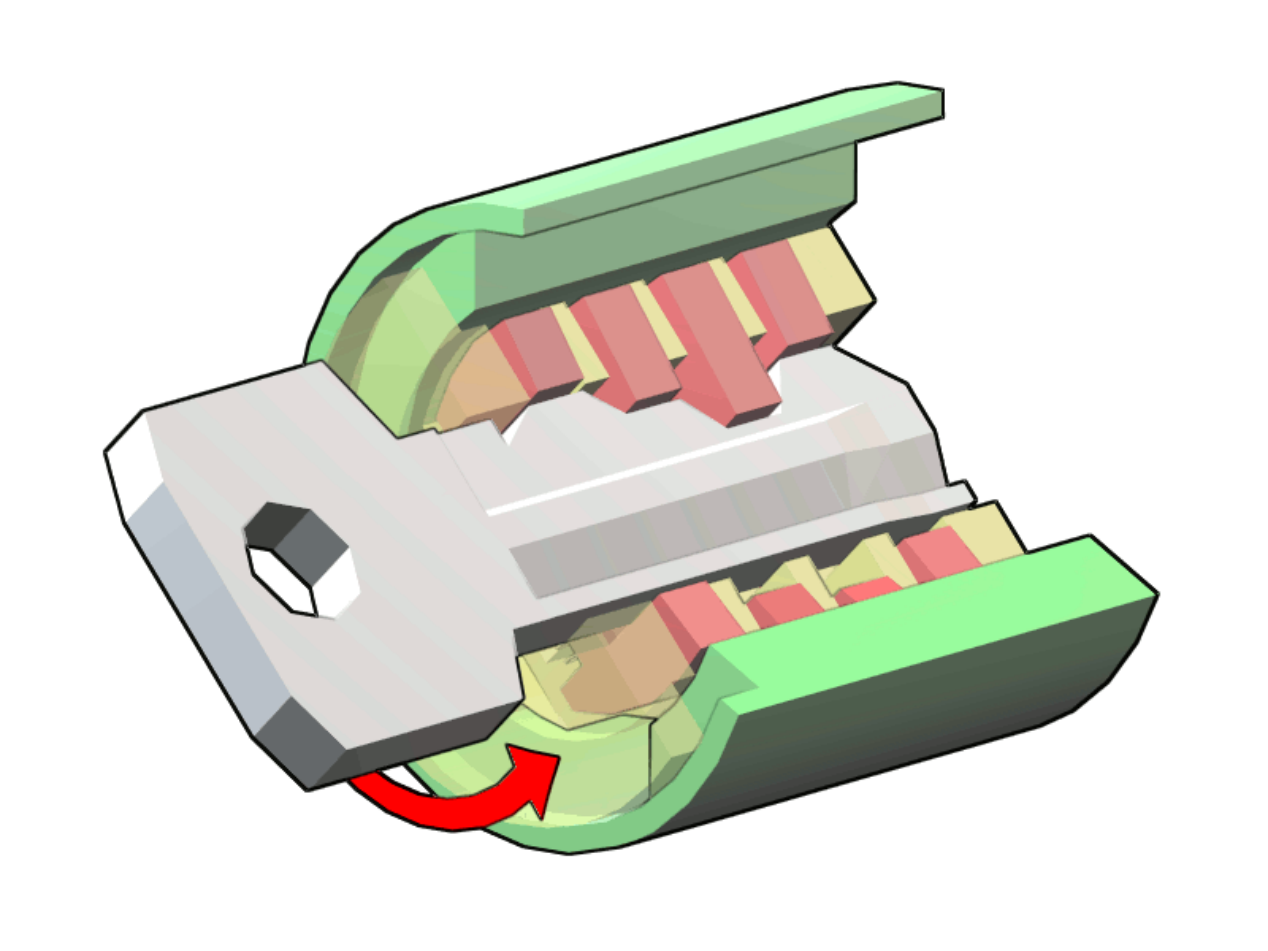
Lever Locks - Used on safe deposit boxes and difficult to pick

Key Lock

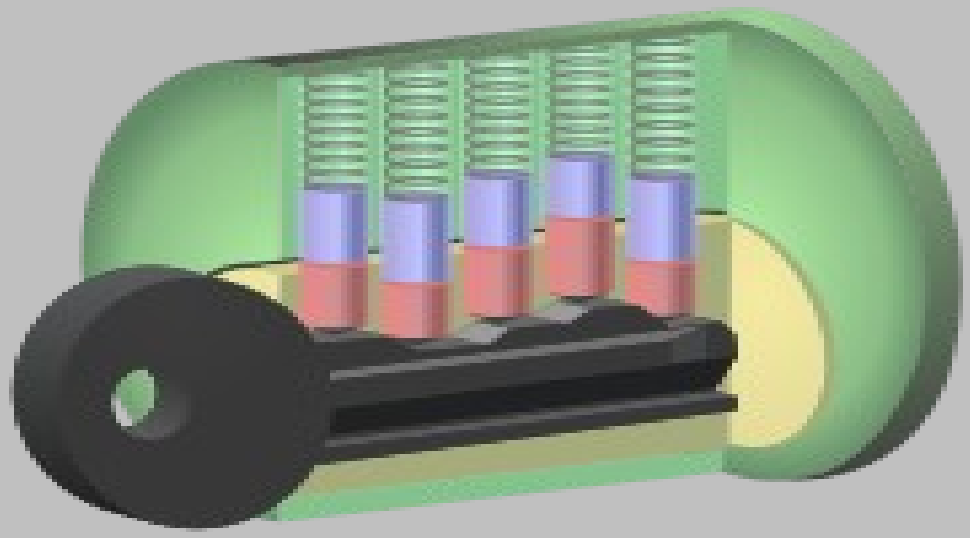
Warded Locks



Disc Tumbler Locks



Pin Tumbler Locks



The **pin tumbler lock** (or **Yale lock**, after lock manufacturers Yale) is a lock mechanism that uses pins of varying lengths to prevent the lock from opening without the correct key.

Conventional Combination Lock - May be opened by skillful manipulator.

Manipulation-Resistant Combination Locks - Provide a high degree of protection for highly classified or important material. Locks with four or more tumblers are desirable for highly important material.

Relocking Devices – Attached to safes or vaults and add additional degree of security against forcible attack.

Interchangeable Core - Cores can be quickly replaced. Locks can be keyed to an overall complete locking system. They are economical, flexible and simplify record keeping.

Cypher Lock - is a digital combination door lock device e.g. X-07. This lock is the first improvement on a dial lock technology since the Civil War.

It is the most technologically advanced combination lock available. It is more secure than mechanical locks and user friendly. Lock is self powered by turning the dial. Audit features on opening and closing, and is now the only lock that meets Federal Spec for classified documents.

Code-operated Locks - Open by pressing a series of number buttons in correct sequence. High security devices

Card-operated Locks - Either electrical or electromagnetic; coded cards which are etched, embossed or contain a pattern of copper flecks are used to operate the locks.

Electro-magnetic Locks - Operate by holding a door closed by magnetism.

Padlocks - Should be hardened and strong enough to resist prying, and should be locked at all times even when not securing an area.

Keying Systems

1. **Change Key** - A key to a single lock within a master key system

2. **Sub-Master Key** - A key which will open all locks within a particular area in a facility.

3. **Master Key** - A key that will open all the locks when two or more sub-master systems exist.

4. **Grand-Master Key** - A key that will open everything in a system involving two or more master key systems.

Key Control

1. Keys should be accessible only to those whose official duty require access.
2. Combination to safes, locks, and padlocks securing containers for classified information should be changed at least once every 12 months or earlier where there has been an actual or possible loss or compromise and when someone knowing the combination has been terminated, left on transfer, or resigned.
3. When selecting combination numbers simple ascending or descending number grouped should be avoided.
4. Storage of combination numbers should be in containers approved for the storage of the same classification of material that the combination is itself protecting.

- Padlocks should be rotated around a facility regularly.
- Inventories of key systems should be carried out annually.
- The use of keys should be based on the general concept as applied to safe combinations.

- When a key is issued the following should be recorded:

    a. Key number
    b. Name of person
    c. Position
    d. Date of issue
    e. Other relevant information

When a key is lost the following actions should be completed.

a. Circumstances should be investigated and a written report produced, and

b. the lock should be changed

Master keys should be kept to a minimum and ideally remain with security.

TRAFFIC CONTROL

1. Traffic control includes the following:

   a. Identification of employees and visitors

   b. Directing movement of employees and visitors.

c. Package control

d. Control of commercial and private vehicles

2. All visitors to the site must be:

- Made to identify themselves
- Be limited to predetermined unrestricted areas
- The most effective method is by ID badges and cards

3. Identification Badges should be:

a. Tamper Resistant

b. Bear a clear and recent photograph of the holder (at least 1”sq. and updated every 2-3 years or when facial appearances change).

c. It must contain personal details such as:

Date of Birth

Height

Weight

Both thumbprints
Color of hair and eyes

d. Should be laminated and of sturdy construction

e. Color coded where necessary

f. Signed by the holder

g. The design should be simple to allow for easy recognition at a distance

3.1 A badge system can only be effective if enforced.

3.2 A package control policy should be in place covering both receipts and dispatches, and widely publicized.

3.3 Incoming packages should be inspected, in most cases a spot check is sufficient.

3.4 Where a practical vehicle parking areas should be away from the facility and protected from the intruders.

3.5 Employees and visitors when leaving the site should pass a check and control point manned by security.

SECURITY FORCE OPERATIONS (GUARDS)

1. Enforcement is the most vulnerable link in any identification system
2. Primary function of guard force is the protection of life and property.
3. Most important security document is the Security Log (24 hour log).

5. Most important document for guard is Special Orders.

6. Controversial guard operations topic is ‘Contract’ versus ‘Proprietary’ and ‘Armed’ versus ‘Unarmed’.

7. Guard requirements 24 hour post and 5 day week:

Days off	No. of Guards
0	4.2
20	4.5
25	4.6
30	4.7
35	4.8
40	4.9

General Guard Notes

- Effective guards routes are systematically unsystematic
- Guards should not be used as escorts
- Area least likely to be checked by security on vehicles is inside the wheel covers or hub caps.

Advantages of Proprietary Guards

- Generally a higher caliber as wages are usually higher
- They generally provide a better service.
- Can be trained to handle more complex security duties
- There tends to be less turnover of staff

- They are more familiar with the facilities they protect
- They tend to be more loyal to the company.

Disadvantages of Proprietary Guards

- They cost more
- There may be problems of ensuring backup personnel to cover sickness, training holidays etc.

Advantages of Contract Guards

- They are less expensive
- They are convenient to use when needed

c. With regard to administration and personnel, problems are reduced as the hirer is relieved of all the payroll related responsibilities.

d. The contractor assumes full responsibility for scheduling and supervising all guard personnel, and is able to obtain extra guards as the post or circumstances dictates.

e. The contract company usually accept liability in civil suits (See paragraph on Legal Aspects below.)

Disadvantages of Contract Guards

- They lack training, particularly knowledge of the hirer site, and procedures.

- Generally a lower caliber of employee, due to lower wages.
- They have little or no loyalty to the hirer.
- They tend to project a poor image
- The contract guard industry is saddled with a large turnover problem.

Recommendations for pre-employment screening qualifications:

a. *Minimum age is 18 years old*

b. *High School diploma or equivalent written exam*

c. *A separate written exam to determine the candidates ability to understand and perform duties assigned.*

d. No criminal convictions.

e. No physical or mental defects that would hinder job performance

f. Be morally responsible in the judgment of the regulatory board

For armed personnel they should meet all the above and in addition:

a. Have 20/20 correctable vision in both eyes

b. Capable of hearing ordinary conversations at 10' with each ear

c. Have no felony convictions involving the use of a weapon

d. have no felony or misdemeanor convictions that would reflect on the applicants ability to perform a security function in an armed capacity.

e. successfully pass a 24 hours firearms course, that includes police and legal requirements, or submit evidence of competency and proficiency; and be required to re-qualify every 12 months while remaining in an armed security guard position.

Numbers of Security Officer

- The number of security personnel required is generally proportional to the size of the facility to be protected.

- The number required to cover a single post around the clock providing coverage for three eight-hour shifts is 4.5 to 5 persons. This allows for vacations, sick leave, etc.

DOGS (K9)

Basic Instincts

a. Hunting, attack without fear

b. Pack, sees human as a member of his pack and will protect him

Primary Value

a. Sensor mechanism, 100 times better than man

b. Hearing 2x that of man

Secondary Value

Attack/ pursue, attack/hold

Dog's Senses

- Smell
- Hearing

c. Sight

d. Touch

Deployment

a. Without handlers

b. With handlers

SECURITY SURVEYS

Security Surveys' are often called "Risk Analysis Surveys" or "Risk Assessment Surveys".

- A Security Survey is defined as the process of conducting an exhaustive physical and thorough examination of all operational systems and procedures of a facility for the following purposes :

a. To determine existing state of security

b. Identifying weakness in defenses

c. To determine the degree of protection required

d. To produce recommendations for a total security system.

3. The survey should be undertaken by either a suitably trained staff security personnel, or a fully qualified independent security specialist.

4. No Universal checklist can be applied to all sites for survey purposes as no two facilities are alike.

5. Before commencing a security survey :

- Written authority should be obtained from the site director
- Previous surveys should be reviewed
- An orientation tour should be made
- Photographs should be taken of things which will be difficult to describe in a report. (Only with authority)

6. After completing the survey an immediate review of the findings should be undertaken with the plant supervisor so that urgent deficiencies can be addressed.

7. A follow-up survey should always be conducted to ensure improvements and recommendations have been made.

8. Any survey report including lists of recommendations is incomplete without including a cost benefit analysis, which is -

“ a direct comparison of the cost of the operation of the security unit and all the existing security measures with the amount of the corporate assets saved or recovered as well as reduction of losses caused by injuries and lost production”.

9. No security measure should be recommended which is not cost effective.